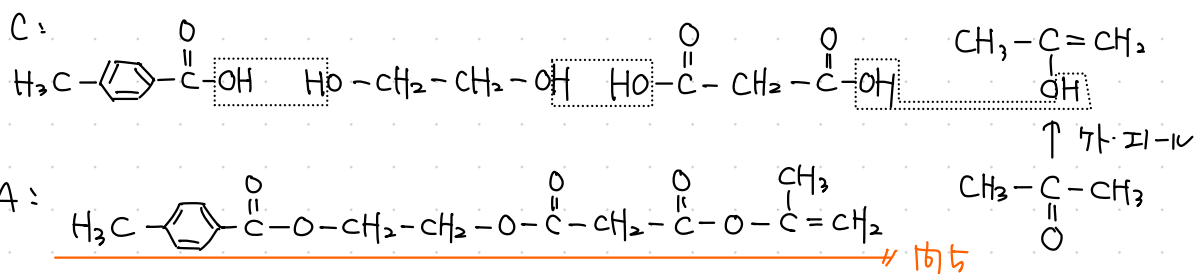
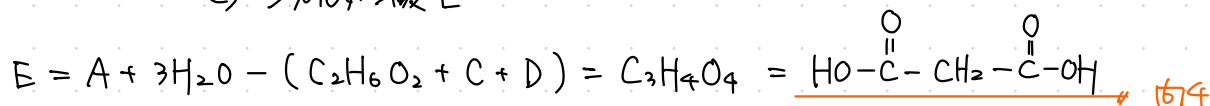
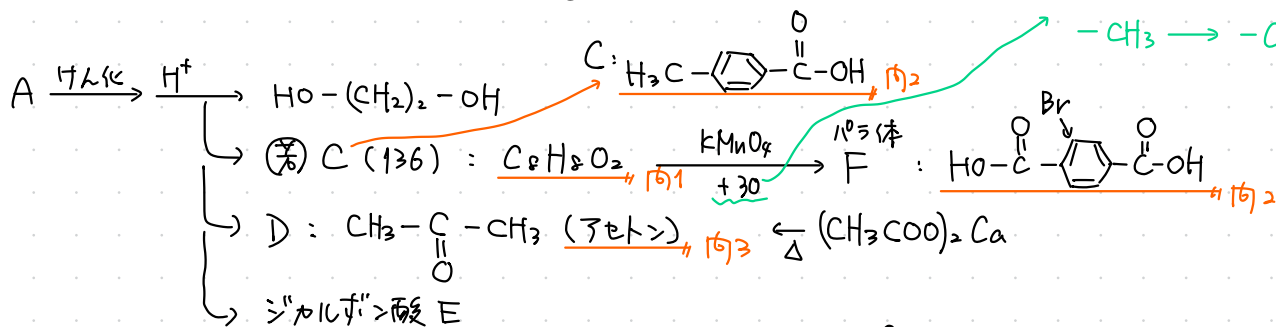


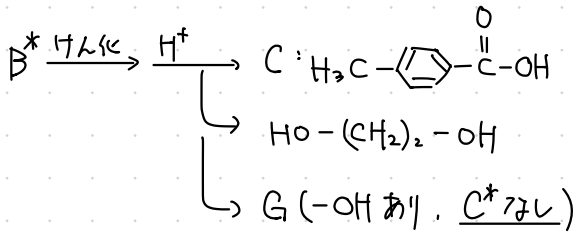
1 THU 2025

エステル A : $C_{16}H_{18}O_6$ $U=8 = -\text{C}(=\text{O})-\text{O}- \times 3 + \text{C}_6\text{H}_5 \times 1 + 1$

$\xrightarrow{\text{KMnO}_4}$ $2^\circ + 30$ は
 $-\text{CH}_3 \rightarrow -\text{COOH}$

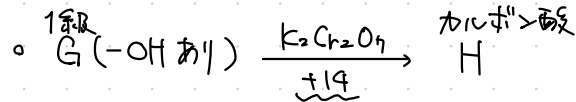


H1エステルB : $C_{16}H_{18}O_6$ $U=8 = -\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}- \times 3 + \text{C}_6\text{H}_4 \times 1 + 1$

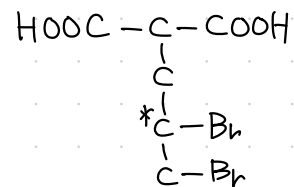
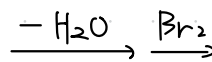
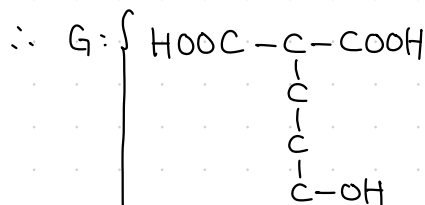
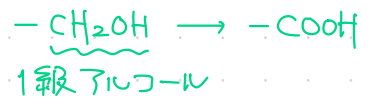
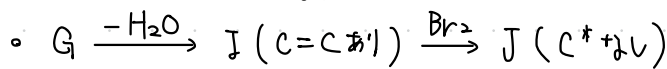


◦ 対称, 面あり

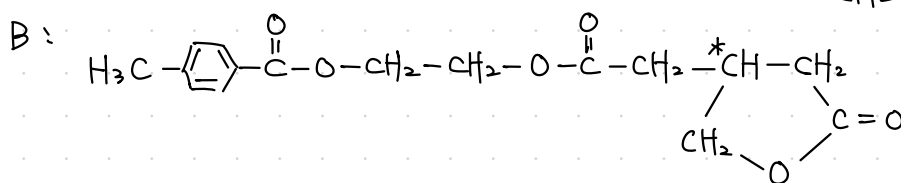
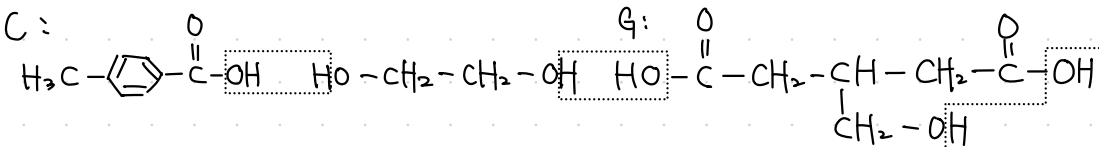
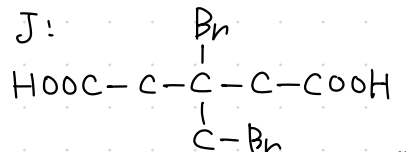
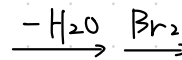
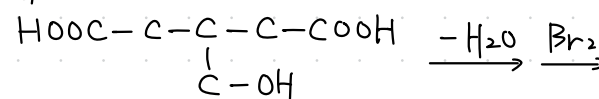
◦ $G = B^* + 3\text{H}_2\text{O} - (C + C_2\text{H}_6\text{O}_2) = C_6\text{H}_{10}\text{O}_5$ ($U=2$)



$\text{K}_2\text{Cr}_2\text{O}_7 \rightarrow \text{Zn} + 14 \text{H}^+$



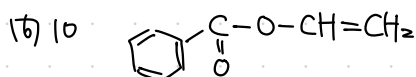
$G:$



167

168 b, d

169 田舎



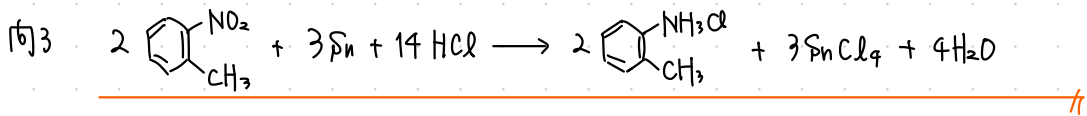
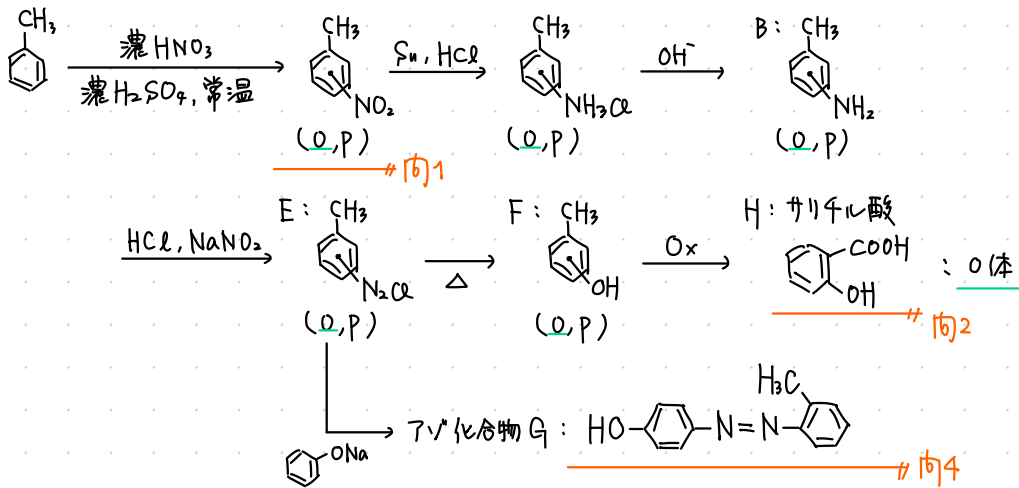
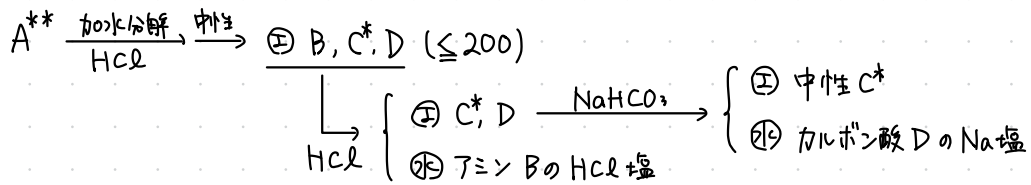
171 2.5 L $\left(\frac{222 \text{ g}}{148 \text{ g/mol}} \times n = 0.60 \text{ mol/L} \times V_L \right)$

172 (1) N: $\text{C}=\text{O} > \left(\text{←} \text{エステル} \right)$ M: PVA

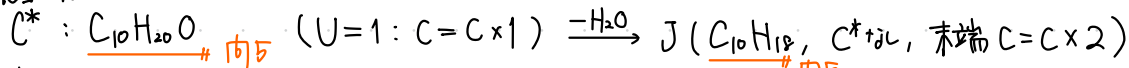
(2) $\frac{22 \text{ g}}{44 \text{ g/mol}} \times (44 + 6 \times 0.32)n \approx 23 \text{ g}$

2 THU 2024

$A^{**} : C_x H_y O_z N_w$, C^* 2017 4級C原子2117個.

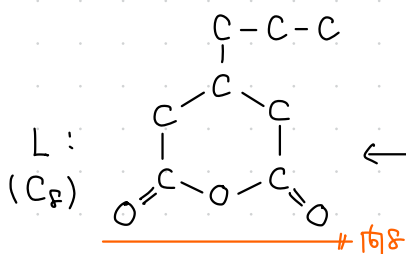
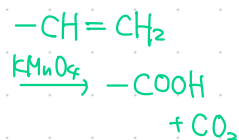
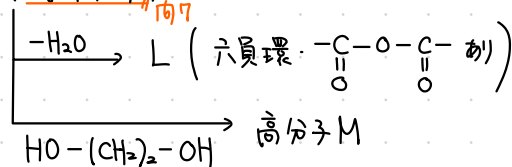
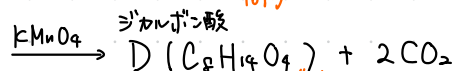


1級
アルコール

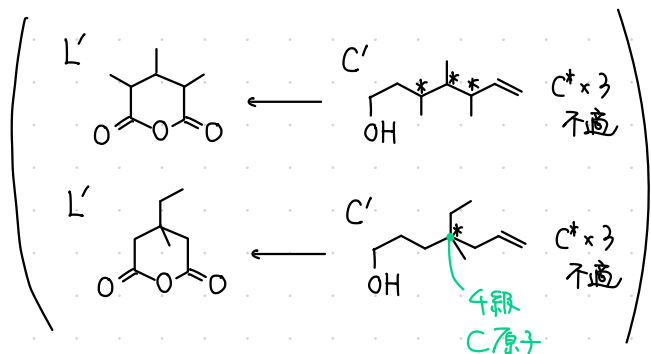
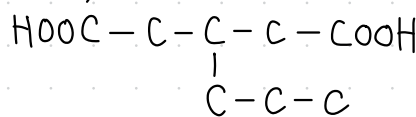


↓ H_2

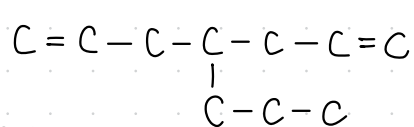
K(C^* なし = 対称面あり)



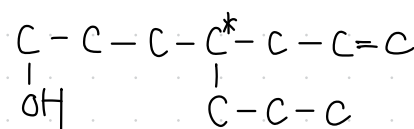
D(C_8)



← J(C_{10})

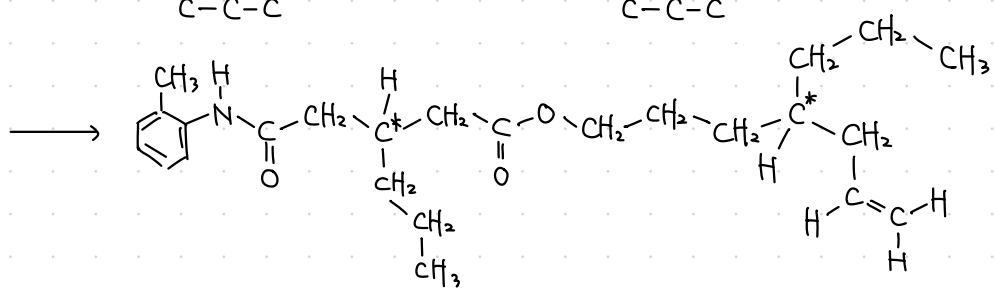
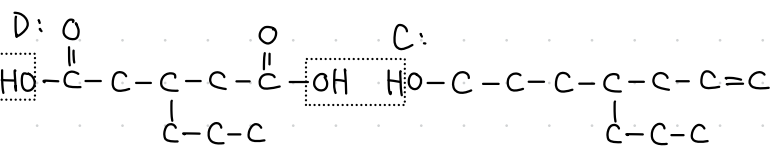
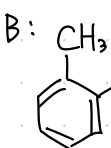
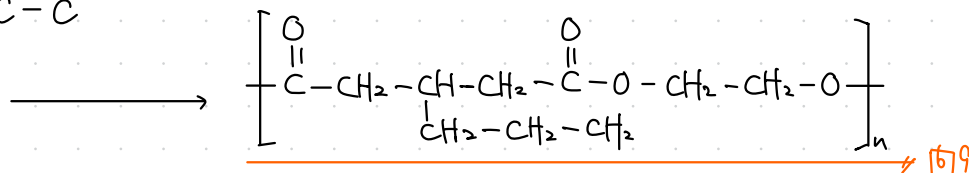
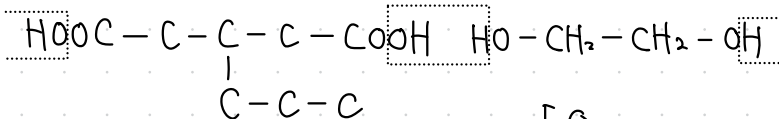


← C(C_{10})



108

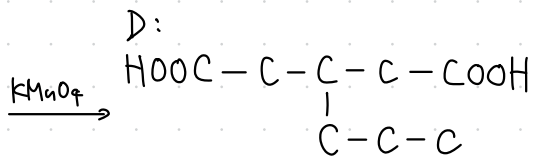
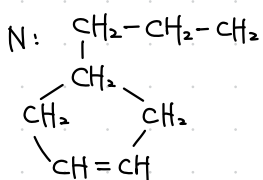
D:



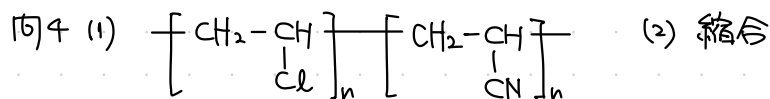
1012

10 2.4 × 10³ g

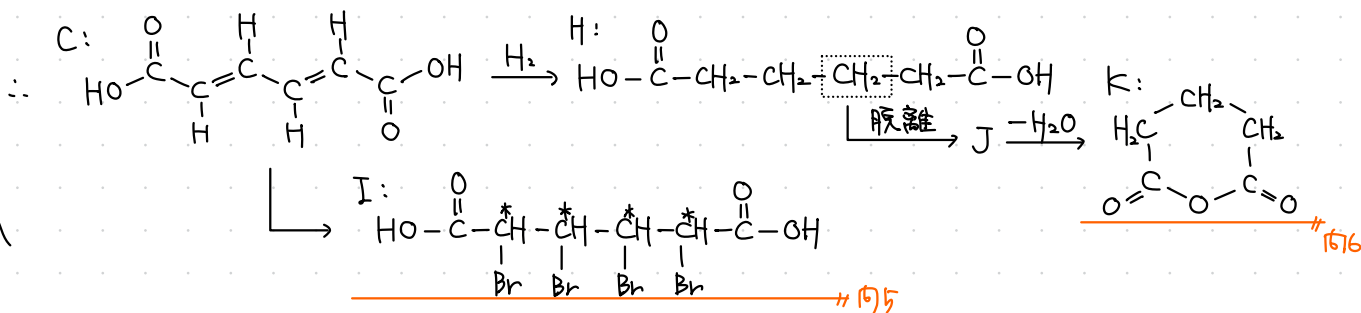
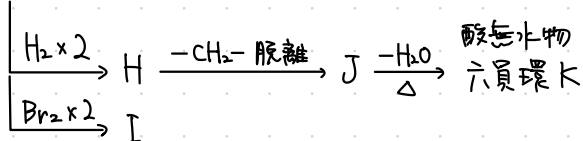
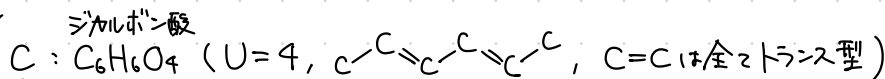
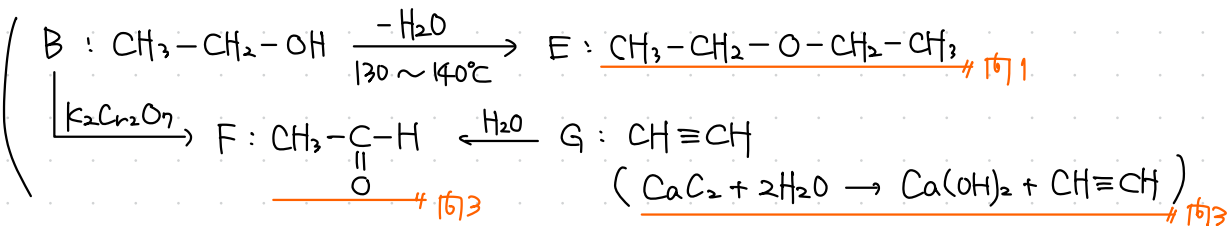
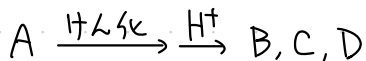
11 C_8H_{14} U=2



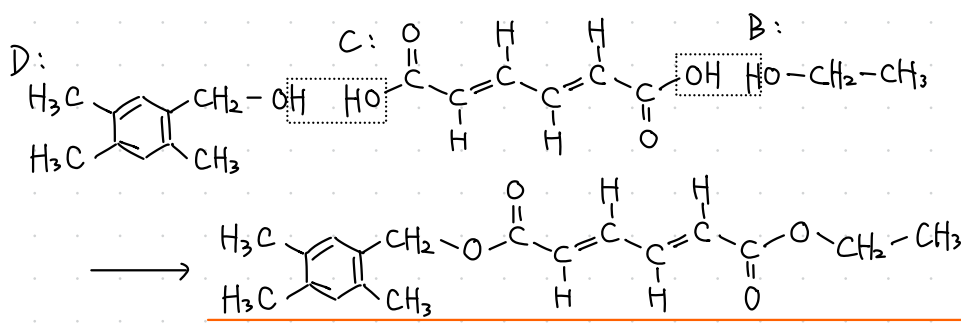
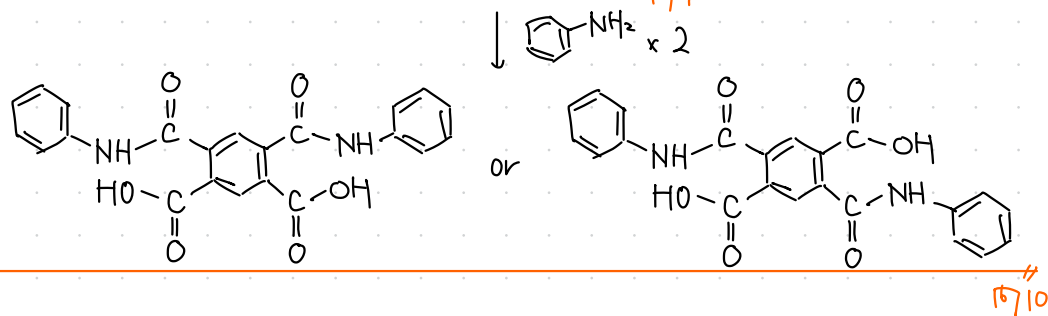
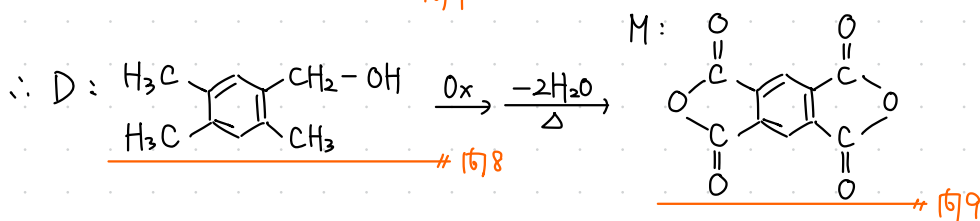
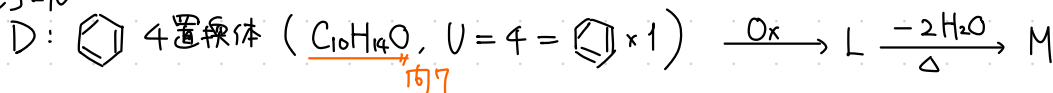
3 THU 2023



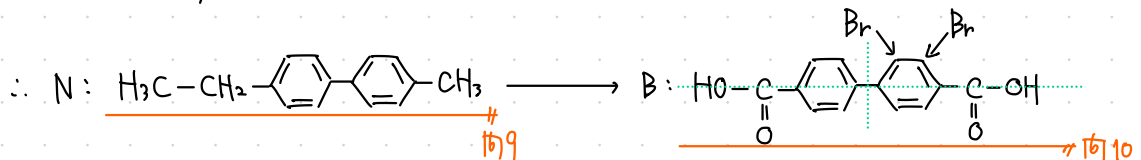
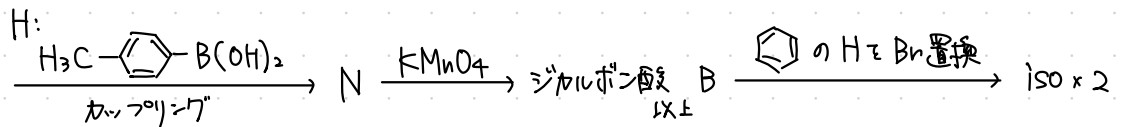
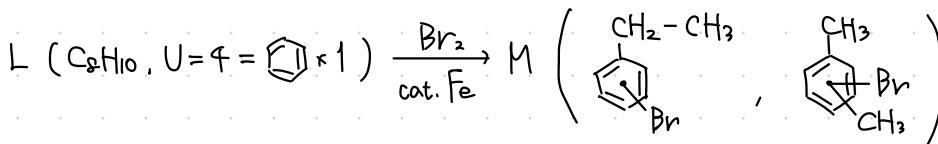
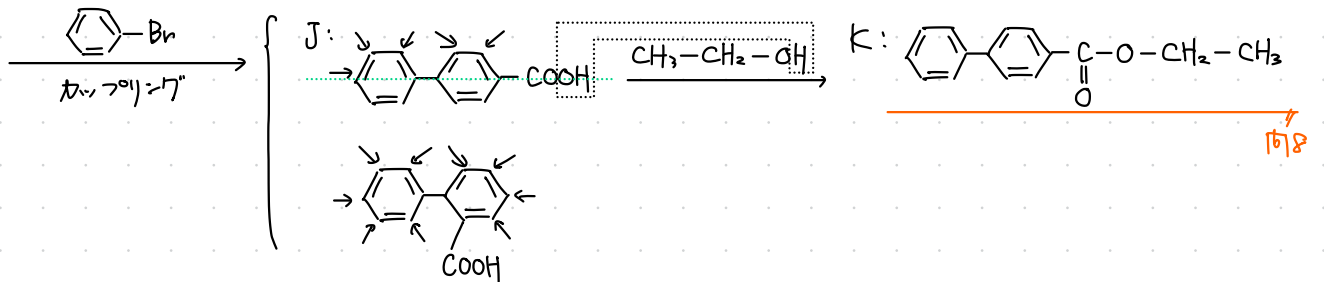
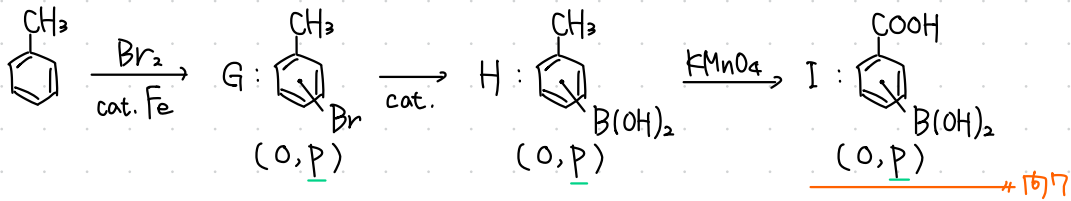
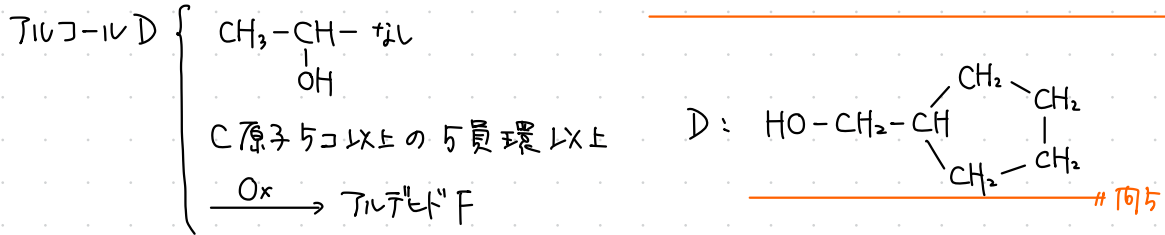
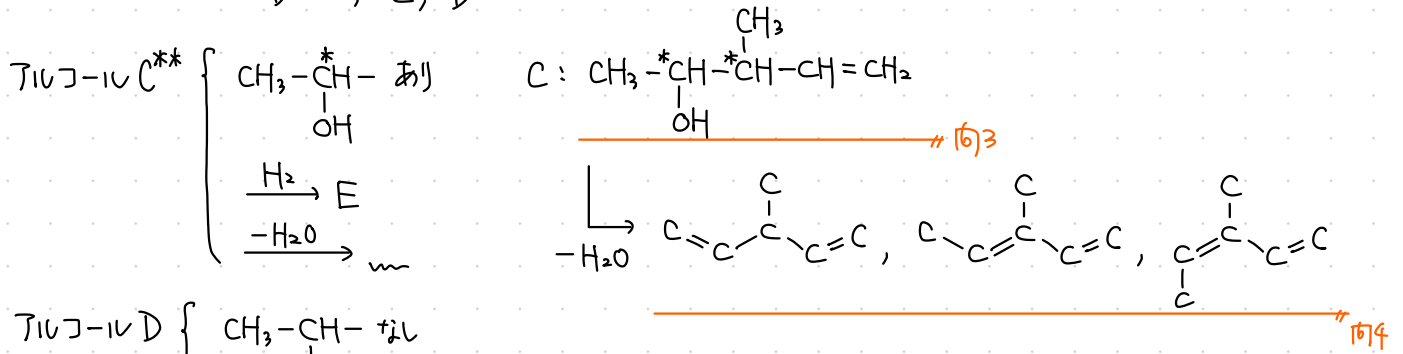
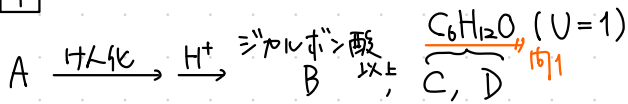
(3) $2.00 \times 10^2 : 1.00 \times 10^2$



第1級
アルコ-ル

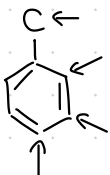


4



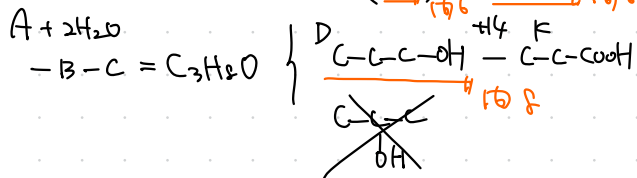
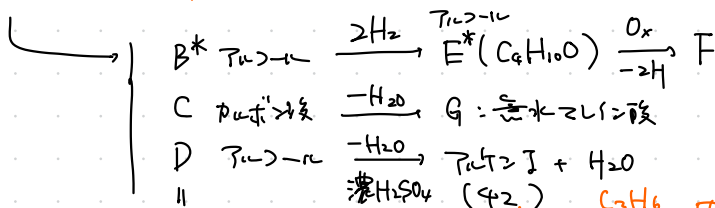
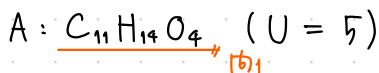
162 田舎

166

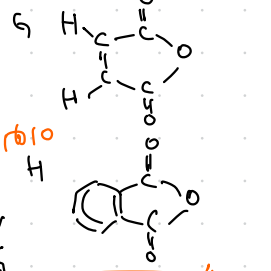
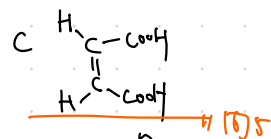
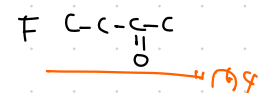
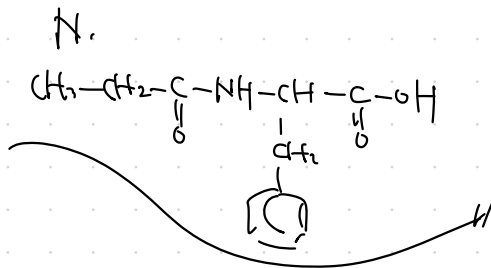


4

5



PP



M: Phe

$N = 74 + 91$
 $-CH_2-$
 $\xrightarrow{H_2} C-C-C$

6

